CHIN-I LIN

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EDUCATION

Washington University in St. Louis, St. Louis, MO | Ph.D. in Neuroscience, 2026
Advisor: Daniel Kerschensteiner, M.D.

National Tsing Hua University, Taiwan | M.S. in Systems Neuroscience, 2018
Advisor: Chuan-Chin Chiao, Ph.D.

National Chung Hsing University, Taiwan | B.S. in Entomology, 2016

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Researcher, Kerschensteiner Laboratory | Washington University in St. Louis | Ph.D. research
- Investigate retinal circuit mechanisms underlying parallel visual processing using calcium/glutamate imaging, electrophysiology, molecular manipulation, and visual behavior.
- Study VGluT3-expressing amacrine-cell dendritic processing and inhibitory control of local receptive fields and contrast-dependent visual responses.
- Investigated the molecular organization and function of the retinal OFF pathway, contributing to a 2025 Nature Communications publication.

Research Assistant, Stroke and Cerebrovascular Diseases Division | Beth Israel Deaconess Medical Center / Harvard Medical School | 2020–2021
- Developed experimental strategies to investigate epigenetic mechanisms in stroke and performed molecular-to-behavioral drug studies.

Research & Development, RNA-NGS Department | Welgene Biotech Co., Ltd., Taipei, Taiwan | 2019–2020
- Developed sample-preparation methods for mRNA sequencing; performed RNA library preparation, quality control, data analysis, and technical reporting.

Research Assistant, Retinal Regeneration Research Group | National Tsing Hua University, Taiwan | 2019
- Used electrophysiological and tissue-culture approaches to study retinal neurite regeneration and contributed to experimental design, analysis, manuscript preparation, and laboratory operations.

PUBLICATIONS

C.-I. Lin, A. Jo, P. Ramakrishna, X. Song, J. L. Morgan, and D. Kerschensteiner. Inhibition confines dendritic integration to set behavior-specific sensory gain. Manuscript in preparation (2026).

F. Soto, **C.-I. Lin**, A. Jo, S.-Y. Chou, E. G. Harding, P. A. Ruzycki, G. K. Seabold, R. S. Petralia, and D. Kerschensteiner. Molecular mechanism establishing the OFF pathway in vision. Nature Communications 16, 3708 (2025).

K. Friedrichsen, J. C. Hsiang, **C.-I. Lin**, L. McCoy, K. Valkova, D. Kerschensteiner, and J. L. Morgan. Subcellular pathways through VGluT3-expressing mouse amacrine cells provide locally tuned object-motion-selective signals in the retina. Nature Communications 15, 2965 (2024).

H. Lu, R. Ashiqueali, **C.-I. Lin**, A. Walchale, V. Clendaniel, R. Matheson, M. Fisher, E. H. Lo, M. Selim, and A. Shehadah. Histone Deacetylase 3 Inhibition Decreases Cerebral Edema and Protects the Blood-Brain Barrier After Stroke. Molecular Neurobiology 60, 235–246 (2023).

H. Chiao, **C.-I. Lin**, and M.-J. Lee. Multiple approaches for enhancing neural activity to promote neurite outgrowth of retinal explants. In Retinal Development, Methods in Molecular Biology, vol. 2092 (2020).

C.-I. Lin and C.-C. Chiao. Blue light promotes neurite outgrowth of retinal explants in postnatal ChR2 mice. eNeuro 6(4) (2019).

CONFERENCE PRESENTATIONS

2024–2026 Inhibition and arbor morphology shape dendritic integration and feature selectivity in the retina. ARVO Annual Meeting, Seattle, WA (2024), Salt Lake City, UT (2025), Denver,CO (2026) .

2018 Blue light promotes neurite outgrowth of retinal explants in postnatal ChR2 mice. ARVO Annual Meeting, Honolulu, HI; 6th Conference on Life Science in Taiwan, Hong Kong and Mainland China; Taiwan Neuroscience Society Annual Conference.

AWARDS & FELLOWSHIPS

2025 Winston Fellow Award, Washington University in St. Louis.

2023–2025 NIH Interdisciplinary Training in Vision Science (T32).

2018 Award for Excellence, 6th Conference on Life Science in Taiwan, Hong Kong and Mainland China.

2017 Honorable Mention Award, NTHU Life Science Graduate Student Symposium.

2016–2017 Research support from BRIM Biotechnology, Inc., for dry-eye research.

2015–2016 Undergraduate Research Award, National Science Council, Taipei, Taiwan.

TEACHING

2023 Mentored Teaching Experience, Principles of Biology, Division of Biology & Biomedical Sciences, Washington University in St. Louis.

2017–2018 Teaching Assistant, Sensation and Perception & Cellular Neuroscience (MOOCs), National Tsing Hua University.

2016–2018 Teaching Assistant, Contemporary Life Sciences & Brain and Mind, National Tsing Hua University.

SELECTED TECHNICAL SKILLS

Imaging: Two-photon calcium/glutamate imaging, confocal microscopy, image acquisition and quantitative image analysis.

Electrophysiology: patch-clamp recording, multielectrode array (MEA) recording.

Retinal & Molecular Techniques: Retina dissection, subretinal/intraocular injection, viral transduction, electroporation, immunohistochemistry, tissue culture, PCR, genotyping, DNA/RNA extraction, RNA library preparation and sequencing QC.

Animal & Behavioral Experiments: Mouse handling, visual behavior assays (looming-evoked defensive behavior, optomotor response), intraperitoneal injection, optic nerve crush.

Programming & Data Analysis: MATLAB, Python; custom analysis pipeline and algorithm development, signal/image processing, statistical analysis, visual stimulus design and experimental control.

Software & Visualization: Fiji/ImageJ, NeuroAnatomy plugin, DeepLabCut, Adobe Illustrator, Adobe Premiere Pro.